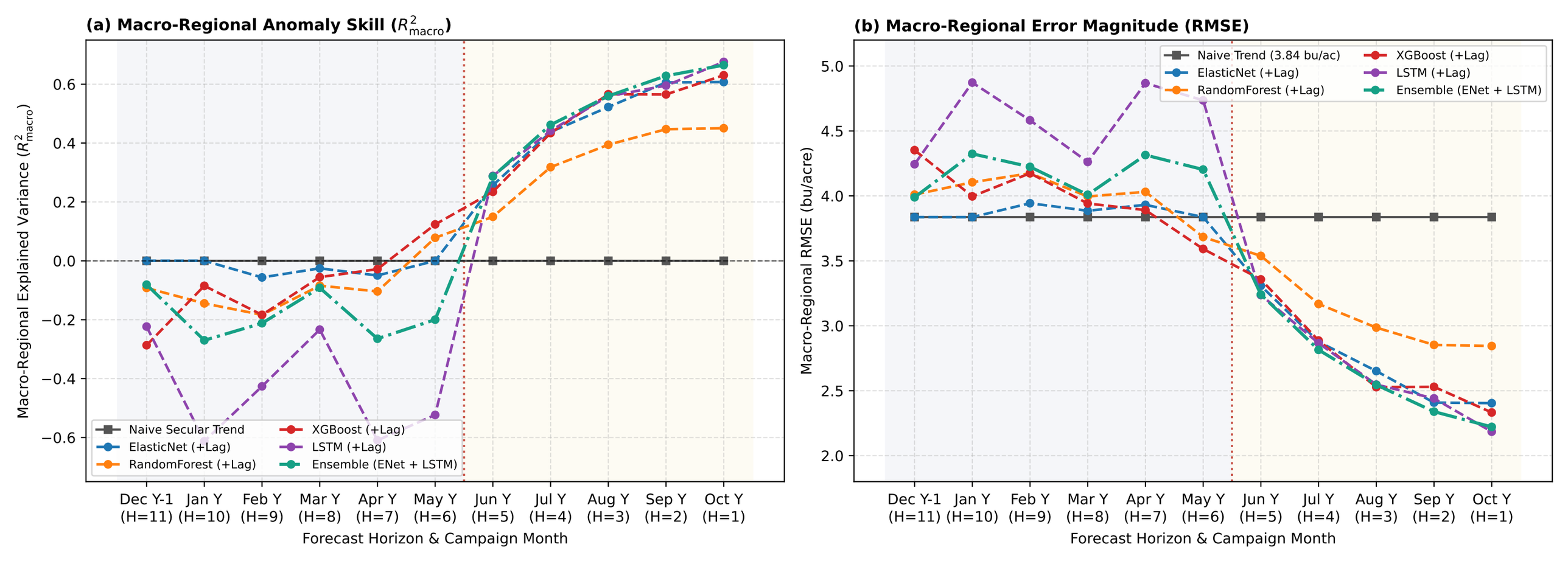
(a) Macro-Regional Anomaly Skill (R^2_{macro})
Macro-Regional Explained Variance (R^2_{macro})
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Naive Secular Trend
ElasticNet (+Lag)
RandomForest (+Lag)
XGBoost (+Lag)
LSTM (+Lag)
Ensemble (ENet + LSTM)
Dec Y-1 (H=11)
Jan Y (H=10)
Feb Y (H=9)
Mar Y (H=8)
Apr Y (H=7)
May Y (H=6)
Jun Y (H=5)
Jul Y (H=4)
Aug Y (H=3)
Sep Y (H=2)
Oct Y (H=1)
Forecast Horizon & Campaign Month
(b) Macro-Regional Error Magnitude (RMSE)
Macro-Regional RMSE (bu/acre)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
Naive Trend (3.84 bu/ac)
ElasticNet (+Lag)
RandomForest (+Lag)
XGBoost (+Lag)
LSTM (+Lag)
Ensemble (ENet + LSTM)
Dec Y-1 (H=11)
Jan Y (H=10)
Feb Y (H=9)
Mar Y (H=8)
Apr Y (H=7)
May Y (H=6)
Jun Y (H=5)
Jul Y (H=4)
Aug Y (H=3)
Sep Y (H=2)
Oct Y (H=1)
Forecast Horizon & Campaign Month